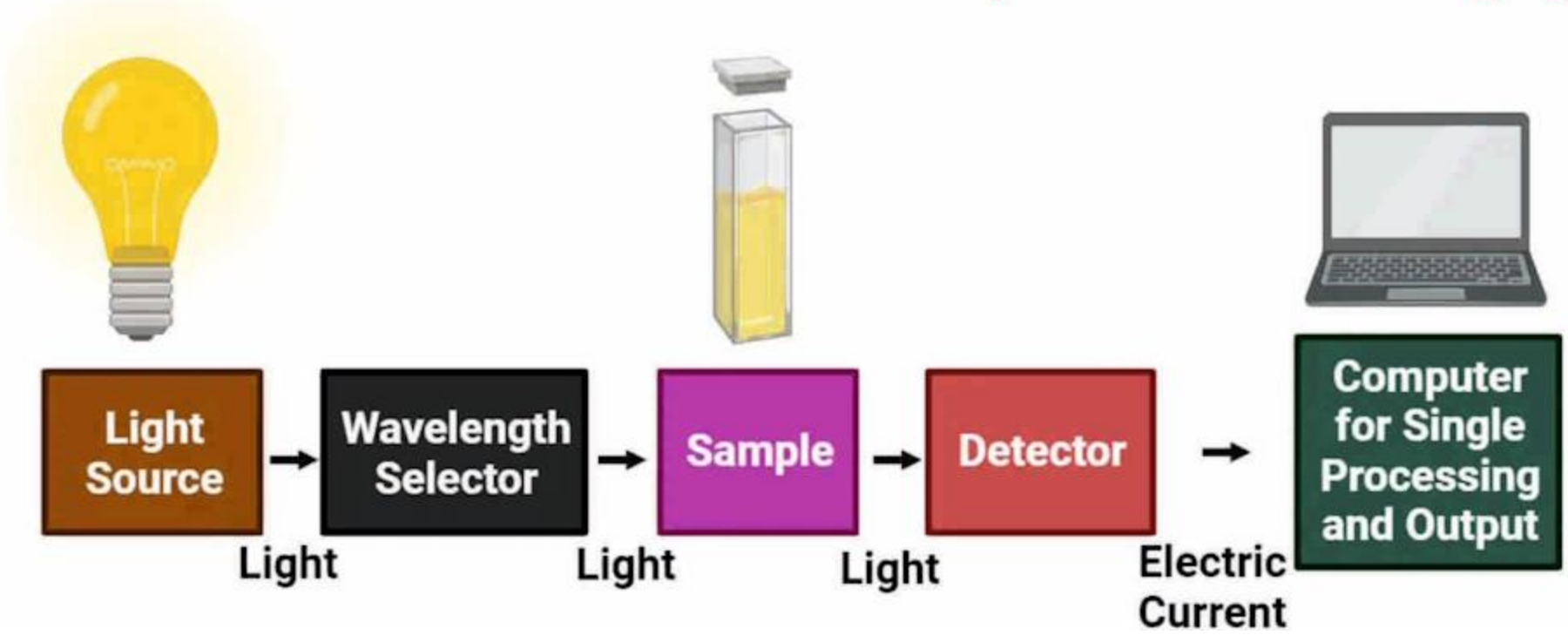
ELECTRONIC SPECTROSCOPY

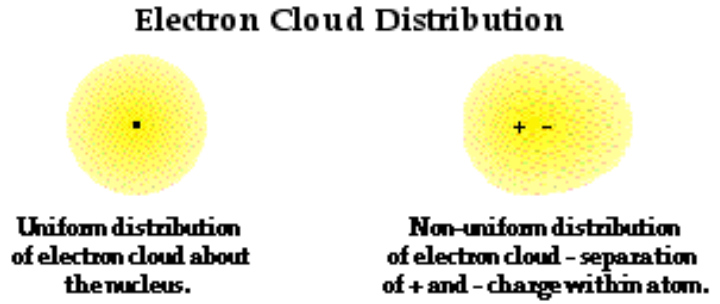
(Ultraviolet and visible spectroscopy)



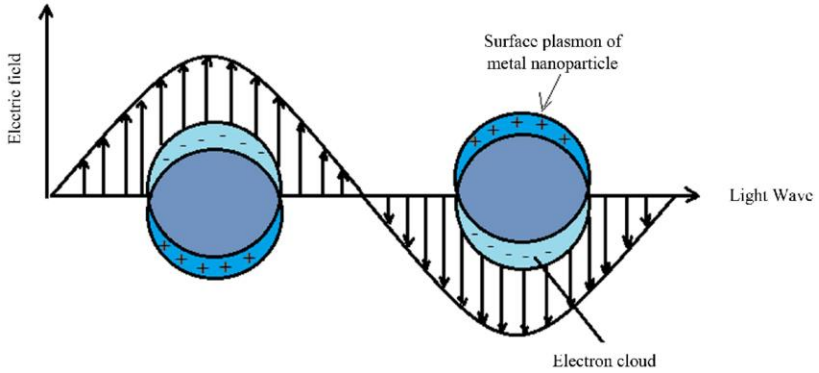
Electric Dipole Transitions

When electromagnetic radiation falls on an atom or a molecule

→ the electric field of the radiation tends to disturb the charge cloud around the atom or the molecule → induces a dipole moment.



→ if the field oscillates between the upper and the lower positions, the induced dipole will also oscillate with the frequency of oscillation of the field → induces the formation of an oscillating dipole.



As long as the frequency of incident radiation ν_i is not close to that of the natural frequency ν_n of the molecule as given by its energy states

$\rightarrow$ the oscillations are due to <u>forced distortion</u> of the molecule by the electromagnetic wave.

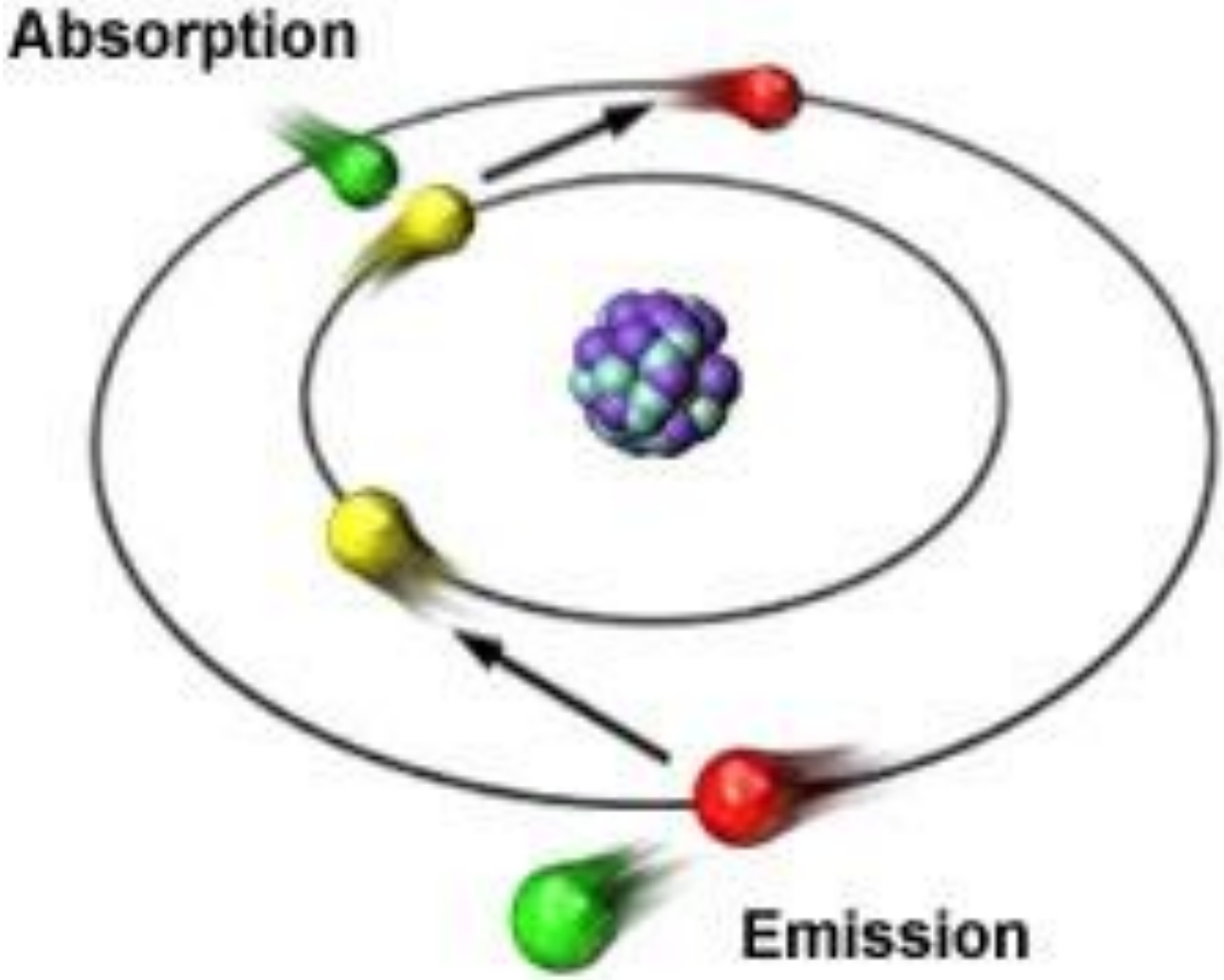
But when $\nu_i \approx \nu_n$, and a **resonance condition** is established between the two interacting partner

$\rightarrow$ the oscillations classically become free.

Under this condition a quantum of radiation or a photon is **absorbed** by the atom or the molecule, promoting it to the higher energy state.

An oscillating dipole moment μ is created and designated as the *transition moment*.

Absorption and Emission of Radiation



Einstein's Treatment of Absorption and Emission Phenomena

According to classical electromagnetic theory

a system of accelerated charged particles emit radiant energy.

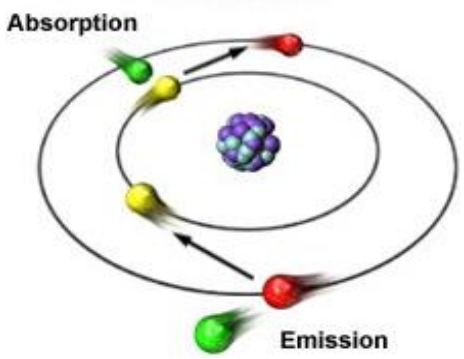
The <u>rates of absorption and emission</u> are given by the classical laws.

These opposing processes are expected to lead to a **state of equilibrium**.

Consider two non-degenerate stationary states *m* and *n* of a system, with energy values E_m and E_n where $E_m > E_n$.

Bohr's theory gives the frequency of the absorbed radiation.

$$\nu_{mn} = \frac{E_m - E_n}{h}$$



Assumption: The system is at lower state *n* and exposed to a radiation of density $\rho(\nu)$ [Energy of radiation per unit volume between the frequencies ν and $\nu + d\nu$].

The probability per unit time that it will absorb the radiation and will thereby raise to the upper state *m*,

is proportional to the number of particles N_n in the state *n* and the density of radiation of frequency ν_{mn}.

$$\text{Rate of absorption} = B_{nm} N_n \rho(\nu_{nm})$$

where, B_{nm} is the proportionality constant and is known as *Einstein's coefficient of absorption*.

The probability of return from *m* to *n* consists of two parts:

(i) Spontaneous (and hence independent of radiation density)
(ii) Proportional to the radiation density.

If N_m be the number of particles in the upper state *m* at any time *t*, then

$$\text{Rate of emission} = A_{mn} N_m + B_{mn} N_m \rho(\nu_{mn})$$

where, A_{mn} is known as <u>*Einstein's coefficient of spontaneous emission*</u> and B_{mn} as <u>*Einstein's coefficient of induced emission*</u>.

At **equilibrium**, the **two rates will be equal** and we have,

$$B_{nm} N_n \rho(\nu_{nm}) = A_{mn} N_m + B_{mn} N_m \rho(\nu_{mn})$$

On rearranging we get,

$$\frac{N_n}{N_m} = \frac{A_{mn} + B_{mn}\rho(\nu_{mn})}{B_{nm}\rho(\nu_{nm})}$$

Boltzmann distribution gives,

$$\frac{N_n}{N_m} = e^{-(E_n - E_m)/kT} = e^{h\nu/kT}$$

$$\frac{N_n}{N_m} = \frac{A_{mn} + B_{mn}\rho(\nu_{mn})}{B_{nm}\rho(\nu_{nm})}$$

$$\frac{N_n}{N_m} = e^{-(E_n - E_m)/kT} = e^{h\nu/kT}$$

Solving for $\rho(\nu_{nm})$ we get,

$$\rho(\nu_{nm}) = \frac{A_{mn}}{B_{nm}e^{h\nu/kT} - B_{mn}}$$

The expression for energy density was given by Planck from his explanations on **Black body radiation**.

$$\rho(\nu_{nm}) = \frac{8\pi h\nu^3}{c^3}\left(\frac{1}{e^{h\nu/kT} - 1}\right)$$

Comparing, we find,

$$B_{n\to m} = B_{m\to n}$$

$$A_{m\to n} = \frac{8\pi h\nu^3}{c^3} B_{m\to n}$$

The expressions for B_{mn} (B_{nm}) and hence A_{mn} can be obtained by applying **time-dependent perturbation theory** in Schrödinger equation.

$$B_{nm} = \frac{8\pi^3}{3h^2}\left[\mu_{xnm}^2 + \mu_{ynm}^2 + \mu_{znm}^2\right]$$

where, the μ's are the components of the *transition moment integral.*

$$A_{mn} = \frac{8\pi h \nu_{mn}^3}{c^3} B_{mn} = \frac{64\pi^4 \nu_{mn}^3}{3hc^3}\left[\mu_{xmn}^2 + \mu_{ymn}^2 + \mu_{zmn}^2\right]$$

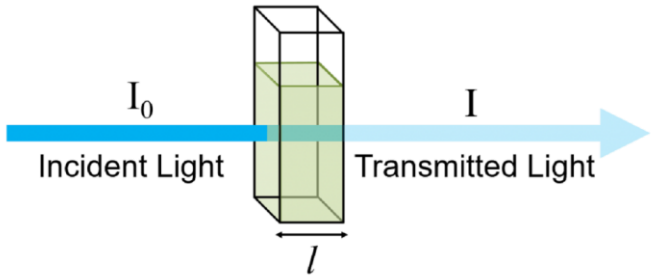
$$= \frac{64\pi^4 \nu_{mn}^3}{3hc^3}\left|M_{mn}\right|^2$$

Square of the transition moment integral

The Lambert-Beer law

When the radiation of frequency (ν) is incident on an absorbing system composed of atoms or molecules, the fractional decrease in intensity on passage through the solution of molar concentration C and layer thickness dl can be expressed as:

$$-\frac{dI}{I} = \alpha_\nu C dl$$



α_ν is the proportionality constant and is a function of incident frequency ν.

On integration over the path length l the expression becomes: $\ln\frac{I_0}{I} = \alpha_\nu C l$

α_ν expresses the probability of absorption per unit time per unit concentration per unit length of the optical path.

Converting to logarithm, we get,

$$\varepsilon_\nu = \frac{\alpha_\nu}{2.303}$$

So, $$\varepsilon_\nu = \frac{1}{Cl}\log\frac{I_0}{I}\,\mathrm{mol^{-1}cm^{-1}s^{-1}} = \frac{1}{Cl}\,\text{optical density}$$

For a range of frequencies, $$\int_{\nu_{nm}} \varepsilon_\nu d\nu = \frac{1}{Cl}\int_{\nu_{nm}} \log\frac{I_0}{I}\,d\nu = A$$

where A is the molar integrated intensity over the absorption band.

Molar integrated intensity over the absorption band

$$\varepsilon_\nu = \frac{1}{Cl}\,\text{optical density}$$ Or, $$\varepsilon_\nu = \frac{1}{Cl}\,\text{absorbance}$$

For a range of frequencies,

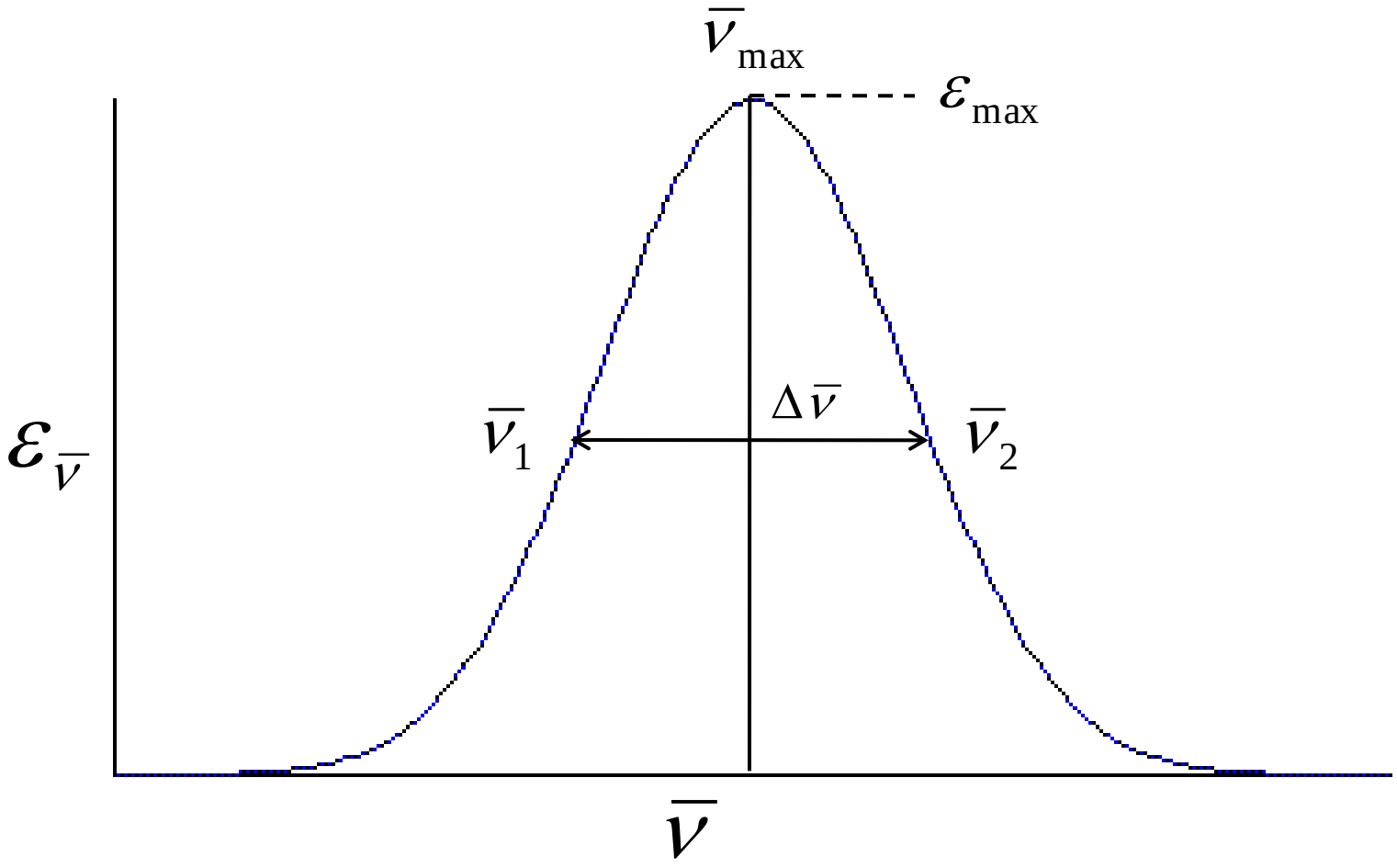
$$\int_{\nu_{nm}} \varepsilon_\nu d\nu = \frac{1}{Cl}\int_{\nu_{nm}} \log\frac{I_0}{I}\,d\nu = A$$ **(Molar integrated intensity over the absorption band)**

In terms of wavenumbers,

$$A = c\int_{\bar{\nu}_{nm}} \varepsilon_{\bar{\nu}} d\bar{\nu} = c\bar{A} \text{ and } \bar{A} = \frac{A}{c}\,ltmol^{-1}cm^{-2}$$

Absorption curve

Area under the absorption band = $\int \varepsilon_{\bar{\nu}} d\bar{\nu} \approx \varepsilon_{\max} \Delta\bar{\nu}$



Molar integrated intensity over the absorption band

The intensity of absorption for an electronic transition is the probability of absorption between two given energy states.

The transition moment integral M_{01} between the ground and the first excited state is expressed as:

$$|M_{01}| = \int \Psi_0 \hat{\mu} \Psi_1 d\tau$$

Where Ψ_0 and Ψ_1 are the wave functions for the two combining states and $\hat{\mu}$ is the dipole moment operator.

Strength of an electronic transition – Oscillator Strength

The strength of an electronic transition is generally expressed in terms of a quantity called *oscillator strength, f.*

It is defined as the ratio of the experimental transition probability to that of the ideal case of a harmonic oscillator.

$$f = \frac{\left[\int \varepsilon_{\bar{\nu}} d\bar{\nu}\right]_{\text{expt}}}{\left[\int \varepsilon_{\bar{\nu}} d\bar{\nu}\right]_{\text{ideal}}} = \frac{\int \varepsilon_{\bar{\nu}} d\bar{\nu}}{2.31\times10^{8}}$$

$$f = 4.33\times10^{-9}\int \varepsilon_{\bar{\nu}} d\bar{\nu}$$

A transition with *f = 1* is known as totally *allowed* transition.

But the transitions between all the electronic, vibrational or rotational states are not equally permitted.

Lifetimes of Excited Electronic States of Atoms and Molecules

The time an electronically excited electron spends in the higher energy states of an atom or molecule, if left unperturbed by the environment, is known as the natural radiative lifetime of the atom or molecule.

After this time the system spontaneously reverts to its original state by emission of radiation of appropriate energy.

In a system of large number of particles, the rate of decay follows a <u>first order rate law</u> and can be expressed as

$$I = I_0 e^{-kt}$$

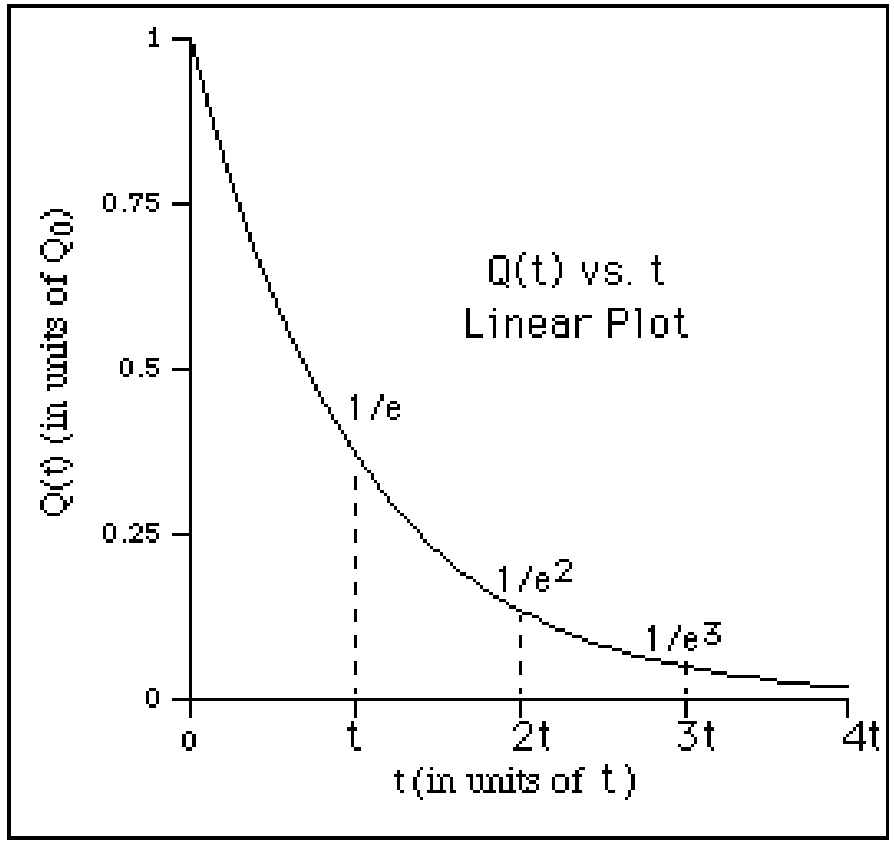
Where I_0 and I are intensities of emitted radiation, respectively, at zero time and at time *t* after the exciting radiation is cut-off; *k* is the rate constant for the emission process and has the dimension of reciprocal time.

If k = 1/τ then $I = I_0 e^{-t/\tau}$

And for a time t = τ $\quad I = I_0 e^{-1} = I_0 \dfrac{1}{e}$

$$I = I_0 e^{-1} = I_0 \frac{1}{e}$$

Therefore the lifetime is defined as the time taken for the radiation intensity to decay to *1/e* th of its original value.

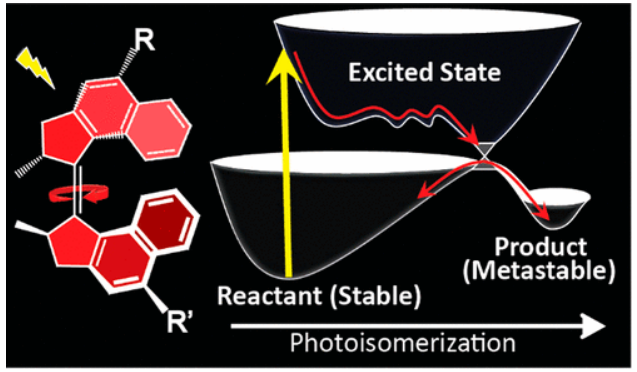


Physicochemical properties of the electronically excited molecules

A molecule in the electronically excited state can be a completely different chemical species with its own wave function and nuclear geometry.

Certain physical properties like dipole moment, pK values, redox potentials differ in comparison to the ground state values.

Excited states, in general, have less deep minima in their potential energy surfaces, indicative of the weakening of attractive interactions.



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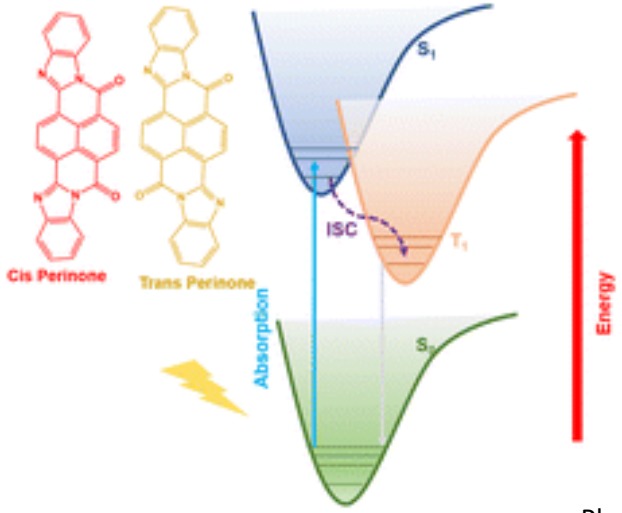

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